A Mamba-Aware Spatial–Spectral Cross-Modal Network for Remote Sensing Classification

Mengru Ma, Jiaxuan Zhao, *Graduate Student Member, IEEE*, Wenping Ma, *Senior Member, IEEE*, Licheng Jiao, *Fellow, IEEE*, Lingling Li, *Senior Member, IEEE*, Xu Liu, *Senior Member, IEEE*, Fang Liu, *Senior Member, IEEE*, and Shuyuan Yang, *Senior Member, IEEE*

***Abstract*— This study introduces a novel cross-modal spatial–spectral interaction Mamba (CMS^{2}I-Mamba) for remote sensing image fusion classification. Unlike convolution-based models focusing on local details and Transformer-based models with high computational complexity, CMS^{2}I-Mamba efficiently models global long-range dependencies in a linear complexity manner. First, multispectral (MS) and panchromatic (PAN) images each have unique advantages in the spectral and spatial attributes. Given this, this article innovatively designs the multi-path selective-scan mechanism (MPS^{2}M), which applies different path scanning strategies to deeply capture the global features from both spectral and spatial dimensions, aiming to enhance the robustness and complementarity of spatial–spectral features. Second, to overcome the characterization differences between images acquired by different sensors, this article further introduces the channel interaction alignment module (CIAM). This module employs efficient former-last and odd-even channel interaction strategies to achieve precise semantic alignment of deep features between modalities. Finally, to leverage the shared fusion features to guide the unique singular features, this article proposes a semantic-aware calibration module (SACM), which accurately constraints and calibrates the same semantic information in deep features. This not only enhances the model's ability to understand scene semantics, but also promotes the deep fusion and utilization of information between different modalities. Through experimental verification on multiple datasets, the CMS^{2}I-Mamba proposed in this article shows excellent recognition performance and computational efficiency (parameter quantity and running speed) in fusion classification tasks. The code for CMS^{2}I-Mamba is available at: https://github.com/ru-willow/CMSI-Mamba.**

***Index Terms*— Fusion classification, Mamba, multimodal, remote sensing, spatial–spectral representation.**

I. Introduction

With the rapid development of satellite technology, the demand for high-quality, high-resolution multispectral remote sensing images is becoming increasingly urgent [1], [2], [3]. However, due to the technical bottleneck of physical sensors, it is still challenging to directly capture images with both high spatial resolution and multispectral characteristics [4]. To this end, modern satellites cleverly integrate two core components: multispectral (MS) and panchromatic (PAN) sensors [5]. The former provides low-resolution but spectrally rich multispectral (LRMS) images, while the latter focuses on capturing high-resolution images with rich texture details [6]. The deep learning-based model seamlessly integrates the high-resolution texture features of PAN images with the multispectral information of LRMS images, obtaining high-resolution multispectral images with high spatial resolution and full spectral characteristics [7]. This process cleverly combines the complementary advantages of the two images, which not only greatly enhances the value and application potential of remote sensing data, but also provides more accurate and comprehensive information support for many fields such as environmental monitoring, resource exploration, and urban planning [8], [9].

In the field of multimodal remote sensing image fusion interpretation, with the development of technology, fusion classification methods based on convolutional neural network (CNN) and Transformer have gradually shown their unique advantages and limitations [10], [11], [12]. CNN has effectively improved the accuracy of object classification through its powerful local feature extraction capabilities and multi-scale fusion strategies, such as the introduction of feature pyramid networks (FPNs) [13]. Its local connection and parameter-sharing mechanisms not only reduce the demand for computing resources, but also enhance the generalization ability of the model. In contrast, the Transformer model [14], [15], [16] can flexibly model long-distance dependencies in sequence data with its self-attention mechanism. It allows the model to capture cross-regional correlation information when processing high-resolution images in image fusion classification. In addition, the parallel computing capability and versatility of the Transformer also facilitate its application in multiple fields.

For fusion classification models based on CNN [17], the semantic nonredundant continuous-scale feature network (SNCF-Net) effectively reduces the information redundancy between bimodal deep features by integrating the texture-enhanced continuous scale input generation module and the cross-modal feature kernel interaction module, thereby

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, International Research Center for Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: lchjiao@mail.xidian.edu.cn).

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strengthening the mapping ability between bimodal features and categories [18]. At the same time, Zhu et al. [19] proposed a multiscale frequency fusion network based on ConvGRU, which cleverly combined the contourlet transform to extract fine texture information, and introduced the inverse band integrator module to achieve deep integration of multiscale enhanced features. This not only enhances the complementary effect between features, but also further optimizes the feature representation through frequency domain operations. Faced with the semantic gap challenge in multisource high-level feature fusion, Ma et al. [20] innovatively designed a cross-source interactive enhancement fusion module, which deeply explored and refined the dependencies between intra-source features, thereby promoting the deep fusion and enhancement of inter-source features. In addition, MBSI-Net [5] built a balanced self-learning system under the framework of a student–teacher network by integrating a texture feature equalization module and a feature fusion module. It not only ensures the balance of feature representation, but also continuously optimizes the fusion effect through a self-learning mechanism. However, their abilities to capture global information are relatively limited and are highly dependent on the quantity and quality of labeled data, which to some extent limits the model's ability to understand and analyze complex scenes [21].

For fusion classification models based on Transformer [22], Xu et al. [23] proposed a dual-stream Transformer with diff-attention (DSTD-Net), which reduced the essential redundancy in features and strengthened the dominant features that are crucial in classification tasks by introducing the self-attention module enhanced by differential attention and the multistage gate fusion strategy. On the other hand, the STGC-Net [24] innovatively constructed an optional self-attention spectral feature extraction framework, which effectively reduced the redundancy generated by the Transformer when segmenting patches, while accurately capturing the channel sequence information of MS image. In order to make up for the inductive bias of the Transformer, Zhang et al. [25] proposed a deep cross-scale Transformer, which further inputs the multiscale output features of the FPN based on attention aggregation into the multiscale global feature learning module and the dense cross-scale attention module, realizing the deep learning and fusion of multilevel cross-scale features. However, it is worth noting that although the fusion classification model based on Transformer shows great potential in feature extraction and fusion, its high computational cost and optimization difficulty are also challenges that cannot be ignored [26].

Although the above works have made a lot of progress in multimodal remote sensing fusion classification, it still faces a series of key problems and challenges that need to be solved.

1) In multimodal remote sensing image processing, MS and PAN images each carry unique spectral and spatial advantages. However, current algorithms all characterize spatial–spectral features in a single and fixed order, which makes it difficult to flexibly capture the complex and changeable unique advantages between the two images [27]. This not only limits the model's understanding of the deep structure of multimodal images, but also hinders its high performance in complex remote sensing scenarios.
2) Since multimodal remote sensing images come from different sensors, they show significant differences in spatial representation. This will hinder the seamless connection and efficient integration of information in the subsequent fusion classification process, as well as thus affect the accuracy and reliability of the classification results.
3) In the process of extracting multimodal remote sensing images feature using deep neural networks, as the network depth increases, the channel spectral information shows a trend increasing layer by layer, and these deep features often contain richer semantic information [28]. If the targeted enhancement of deep fusion features and potential singular features is ignored, this will greatly limit the performance of the model in pixel-by-pixel classification tasks.

Therefore, in response to the above problems, this article proposes a cross-modal spatial–spectral interaction Mamba (CMS^{2}I-Mamba), which is applied Mamba to the fusion classification of MS and PAN images for the first time. The CMS^{2}I-Mamba not only retains the sensitivity of CNN in capturing local features of images, but also draws on the global vision ability of Transformer. It realizes the deep integration and mutual enhancement of different modal images in spectral and spatial dimensions. The specific contributions are as follows.

1) For the characterization of spectral and spatial features, this article innovatively proposes a multipath selective-scan mechanism (MPS^{2}M), which abandons the limitations of fixed order representation and adopts a flexible and changeable spectral and spatial scanning strategy. It introduces different scanning paths (spectral Mamba and spatial Mamba) according to the characteristics of different modal images, thereby effectively filtering out irrelevant information while maximally retaining and strengthening the necessary and relevant features for the classification task.
2) For the alignment of multimodal latent space, this article designs a channel interaction alignment module (CIAM), which breaks the limitation of the independence of channels in traditional methods, adopts former-last and odd-even channel interaction mechanisms, promotes the flow and fusion of information between different channels, as well as is conducive to more efficient and accurate information integration and semantic alignment.
3) For the representation of deep features, this article designs a semantic-aware calibration module (SACM). It cleverly uses the fused common features as a bridge to perform deep semantic calibration on the unique features from different modalities with their advantageous attributes, aiming to ensure that even in diverse modalities, the unique features can achieve a high degree of consistency at the semantic level and jointly represent the same core information.

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The present article is structured as follows. Section II explains the relevant modules, Section III comprehensively analyzes the implementation details and operation procedures of CMS^{2}I-Mamba, Section IV verifies the effectiveness of CMS^{2}I-Mamba through a series of rigorous experiments, as well as Section V is the conclusion and future development direction of this article.

II. Related Work

A. Structured State Space Models

In the field of deep learning, sequence processing models have evolved gradually from recurrent neural networks (RNNs) to long short-term memory networks (LSTMs), to gated recurrent units (GRUs) [29], and finally to the Transformer [30] model based on the self-attention mechanism. Although these models have achieved remarkable achievements, they each face challenges such as gradient problems, high computational complexity, and high resource consumption when processing long sequences.

Therefore, Mamba [31]: the selective scanning structured state-space sequence model (S6) (see Fig. 1), which originated from classical control theory, has been recently proposed. It provides linear scalability of sequence length for long-range dependency modeling. Furthermore, it relies on a classical continuous system. The system maps a 1-D input function or sequence $x_t \in \mathbb{R}^L$ to an output $y_t \in \mathbb{R}^L$ through an implicit latent state $h_t \in \mathbb{R}^N$. The S6 usually contains state equations and output equations. In discrete-time systems, these equations can be expressed as

$$\begin{aligned} h_t &= \bar{A}h_{t-1} + \bar{B}x_t \\ y_t &= Ch_t + Dx_t \\ s.t. &\begin{cases} \overline{A} = \exp(\Delta A) \\ \overline{B} = (\Delta A)^{-1}(\exp(\Delta A) - I)\Delta B \end{cases} \end{aligned} \tag{1}$$

where Δ represents the time scale parameter. $A \in \mathbb{R}^{N\times N}$ is the state transfer matrix, $B \in \mathbb{R}^{N\times 1}$ is the input matrix, $C \in \mathbb{R}^{N\times 1}$ is the output matrix, and $D \in \mathbb{R}^N$ is the direct transfer matrix (which may not exist in some cases, i.e., $D = 0$). In addition, the above formula can be mathematically equivalently converted into the following CNN form:

$$\begin{aligned} \bar{K} &\triangleq (C\bar{B}, C\overline{AB}, \ldots, C\bar{A}^{L-1}\bar{B}) \\ y &= x * \bar{K} \end{aligned} \tag{2}$$

where L is the sequence length, $*$ represents the convolution operation, and $\bar{K} \in \mathbb{R}^L$ is a structured convolution kernel.

The Mamba model's design concept emphasizes minimizing computational costs and memory requirements as much as possible while maintaining high performance, making it an ideal choice for deployment in resource-constrained environments [32]. It can be applied to remote sensing image classification tasks due to its global modeling capabilities, linear complexity, dynamic multipath activation mechanism, and the coexistence of high efficiency and accuracy. Recently, the multimodal image fusion dynamic feature enhancement FusionMamba [33] proposed by Xie et al. had achieved state-of-the-art performance in various multimodal medical image fusion tasks. However, with the deepening of research, Dong et al. [34] further explored the potential of the Mamba model. They introduced an improved Mamba architecture combined with a gating mechanism to achieve more sophisticated cross-modal fusion in the hidden state space. At the same time, Li et al. [35] focused on the optimization application of the Mamba model in specific tasks and proposed the MambaDFuse, which showed significant performance improvements in downstream tasks such as object detection, verifying the flexibility and effectiveness of the Mamba framework in complex visual tasks. Nevertheless, it is worth noting that existing studies have focused on the optimization and application of a single or specific Mamba model, and rarely explored the interaction mechanism between Mamba models with different attributes. This may limit the model's potential for deeper feature fusion, which in turn affects its performance in complex multimodal data processing tasks.

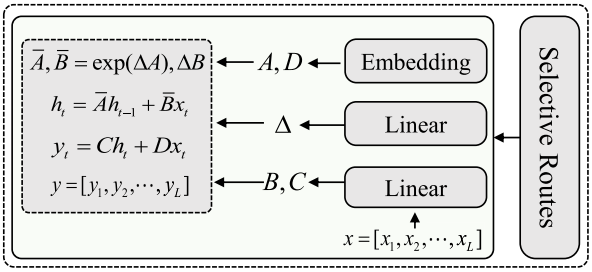


Fig. 1. Mamba: selective scanning structured state space models for sequence models (S6).

B. Multimodal Fusion Classification

Multimodal fusion classification uses a variety of different types of remote sensing data, such as optical images, radar images, and hyperspectral images, combined with deep learning information extraction methods to reflect more comprehensively scene information, thus improving ground object classification accuracy and robustness [36], [37], [38], [39], [40].

For instance, the CRHFF offered by Liu et al. [41] effectively alleviated the adverse effects of local patch effects on the spatial integrity and connectivity of objects by integrating hidden layer features at different resolutions. Furthermore, the AM^{3}Net [42] framework used involution operations to deeply mine spectral information, and combined multiscale mutual guidance learning strategies to generate more accurate and discriminative feature representations. As a deep learning network based on the encoder–decoder architecture, GCFnet [43] innovatively designed two shallow-to-deep and multiscale cross-modal feature fusion branches, as well as effectively integrated the classification advantages of each branch through a probability-weighted decision fusion strategy. In addition, the MFT [44] showed excellent performance in the land cover classification task of hyperspectral and multimodal images by introducing the multihead cross-patch attention (mCrossPA) mechanism. It not only strengthened the

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model's ability to learn long-range feature dependencies, but also promoted the improvement of the model's generalization ability. The deep feature interaction network (DFINet) proposed by Gao et al. [45] cleverly balanced spatial consistency and feature distinguishability under the comprehensive constraints of consistency, discriminability, and classification loss functions. The nearest neighbor-based contrastive learning network (NNCNet) suggested by Wang et al. [46] deeply explored the high-order feature interaction and alignment between multimodal data through an innovative nearest neighbor data enhancement scheme and a bilinear attention module, greatly enriching the semantic information of feature representation. However, these methods do not take into account the diverse representations of different advantages, and lack constraints on the final identical semantic information, which is not conducive to subsequent interpretation tasks.

III. Methodology

In this section, this article introduces in detail the proposed CMS^{2}I-Mamba (see Section III-A), which mainly includes a MPS^{2}M (see Section III-B), CIAM (see Section III-C) and semantic aware calibration module (SACM) (see Section III-D).

A. Cross-Modal Spatial–Spectral Interaction Mamba

To make up for the defects that a CNN primarily focuses on extracting local features, while a Transformer excels at capturing global dependencies but incurs higher computational complexity (the self-attention mechanism in Transformer causes its complexity to be $O(n^2 \cdot d)$, where n is the length of the input sequence and d is the dimension of the vector representation of each token), this article proposes a CMS^{2}I-Mamba (see Fig. 2). Given that Mamba can selectively focus on what must be paid attention to and filter out what can be ignored, as well as has both training and reasoning efficiency, we choose Mamba as the baseline module.

The multispectral image ($MS \in \mathbb{R}^{4\times16\times16}$: with a patch size of 16) provides rich spectral data by capturing the radiation information of multiple spectral bands, making the identification and analysis of land object types more accurate. The panchromatic images (PAN $\in \mathbb{R}^{1\times64\times64}$: with a patch size of 64) focus on high-resolution imaging of a single visible light band, showing the texture and structural characteristics of the surface with extremely high pixel details.

The MS and PAN images are initially processed through a ResNet block (ResBlk) [47] and patch embedding (PE), resulting in feature spaces $\mathrm{MS}_{\mathrm{RB}} \in \mathbb{R}^{128\times64}$ and $\mathrm{PAN}_{\mathrm{PE}} \in \mathbb{R}^{64\times64}$, respectively. Subsequently, $\mathrm{MS}_{\mathrm{RB}}$ is fed into the spectral Mamba, while $\mathrm{PAN}_{\mathrm{PE}}$ is input to the spatial Mamba along with fully connected layers, yielding $F^m_{\mathrm{spe}} \in \mathbb{R}^{128\times64}$ and $F^m_{\mathrm{spa}} \in \mathbb{R}^{128\times64}$. This can model global dependencies in different scanning ways from both channel and spatial dimensions. These outputs are further processed through CIAM to obtain $F^{\mathrm{MS}}_{\mathrm{CIAM}} \in \mathbb{R}^{128\times64}$ and $F^{\mathrm{PAN}}_{\mathrm{CIAM}} \in \mathbb{R}^{128\times64}$. This facilitates semantic alignment in deep feature space through former-last and odd-even channel interactions. In addition, $F^{\mathrm{MS}}_{\mathrm{CIAM}}$ and $F^{\mathrm{PAN}}_{\mathrm{CIAM}}$ are add to generate the fusion feature $F^{Fu}_{\mathrm{CIAM}} \in \mathbb{R}^{128\times64}$. Finally, $F^{\mathrm{MS}}_{\mathrm{CIAM}}$ and F^{Fu}_{CIAM} or $F^{\mathrm{PAN}}_{\mathrm{CIAM}}$ and F^{Fu}_{CIAM} are fed into SACM, producing enhanced semantic perception calibration features $F^{\mathrm{MS}}_{\mathrm{SACM}} \in \mathbb{R}^{128\times64}$ and $F^{\mathrm{PAN}}_{\mathrm{SACM}} \in \mathbb{R}^{128\times64}$, along with their sum F^{Fu}_{SACM} directed into a multilayer perception (MLP: $\mathcal{M}_{lp}(\cdot)$) for subsequent classification

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log\left(\mathcal{M}_{lp}(F^{Fu}_{\mathrm{SACM}})\right)_{ij}. \tag{3}$$

Among them, N is the number of samples in a batch, C represents the number of categories marked in the ground truth, y_{ij} indicates whether the ith sample belongs to the jth category, and its value is 0 or 1. $(\mathcal{M}_{lp}(F^{Fu}_{\mathrm{SACM}}))_{ij}$ stands for the probability that the ith category belongs to the jth category. In Fig. 2, N_1, N_2, N_3, and N_4 represent the number of spectral Mamba, spatial Mamba, CIAM, and SACM modules, respectively. The specific design is detailed in Section IV-B.

The CMS^{2}I-Mamba cleverly combines the rich spectral information of MS images with the excellent spatial resolution characteristics of PAN images, which achieves deep representation of features in both spectral and spatial dimensions through an MPS^{2}M. Furthermore, with the in-depth extraction of semantic information, the CMS^{2}I-Mamba framework effectively reduces the representation gap between features of different modalities through an efficient CIAM. This not only promotes the alignment of features between modalities, but also significantly improves the joint representation capability of features. It is particularly worth mentioning that, in the end, through the fine calibration of the fused features to the unique features, the SACM effectively suppresses the negative impact of singular features on the overall performance, further improving the robustness and generalization ability of the features.

B. Multipath Selective-Scan Mechanism

The fixed-order single-path scanning method in Mamba may not be able to adapt to the changing data sequences in practice, limiting the generalization ability of the model and increasing the risk of overfitting. Therefore, this article proposes a MPS^{2}M (see Fig. 3) for the input sequence. At the same time, given that MS and PAN images have different spectral and spatial advantages, this article designs different multistream selective scanning mechanisms for the spectral domain and spatial domain.

1) Spectral Mamba: Compared with PAN images, MS images contain rich spectral information. Therefore, this article designs a spectral Mamba: a selective scanning structured state space model for spectral sequences. The specific design is shown in Fig. 3(a). This article uses scanning in different directions to expand the image channels to three different directions: forward, reverse, and random, thereby generating three independent sequences. The scanning in these three directions ensures that each channel in the feature map contains information from all other channels in all directions. Therefore, it establishes a comprehensive global receptive field without linearly increasing the computational complexity. Subsequently, Mamba is used to process each feature sequence. The specific design is shown in Fig. 3(c).

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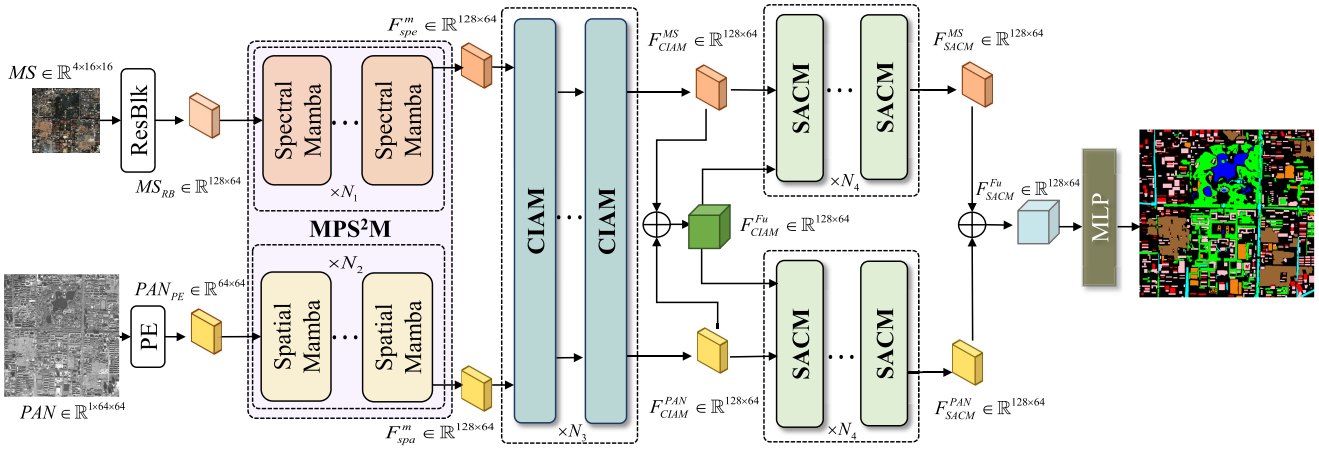


Fig. 2. CMS^{2}I-Mamba. MPS^{2}M is a multipath selective-scan mechanism, CIAM means channel interaction alignment module, and SACM stands for semantic aware calibration module. $\oplus$ means element-wise addition. N_1, N_2, N_3, and N_4 represent the number of spectral Mamba, spatial Mamba, CIAM, and SACM modules, respectively.

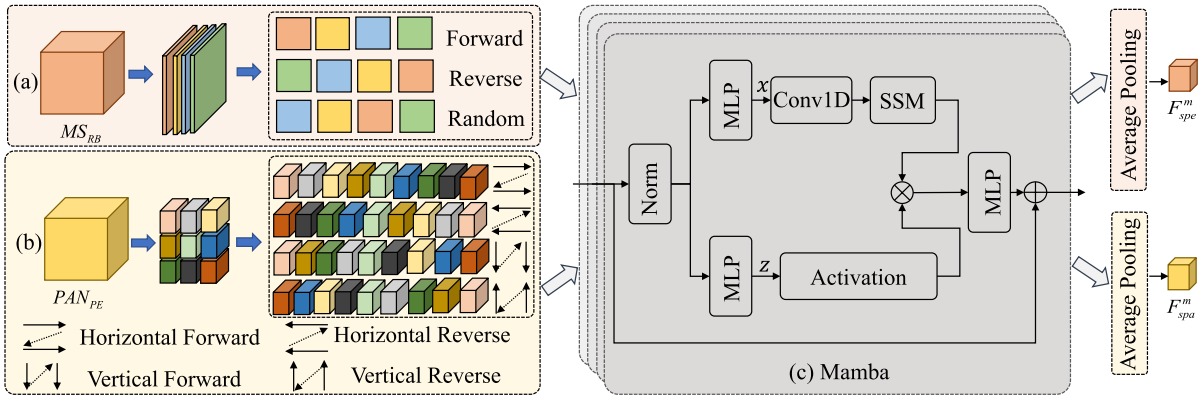


Fig. 3. MPS^{2}M. (a) Spectral Mamba. (b) Means spatial Mamba. (c) Represents Mamba, where the "Activation" function is SiLU.

Taking the forward sequence scan as an example, $f_{\text{fwd}}(MS_{\text{RB}})$ is input into Mamba to model the dependencies between channels. The specific process is as follows:

$$F_{\text{fwd}}^{\text{MS}} = \mathcal{M}_{lp}(\mathcal{S}_{SM}(\mathcal{C}_{\text{onv}}(x)) \otimes \varphi(z)) + F_{\text{fwd}}^{\text{MS}}$$
$$s.t. \quad x, z = \mathcal{M}_{lp}(\mathcal{N}(f_{\text{fwd}}(MS_{\text{RB}}))). \tag{4}$$

Among them, $f_{\text{fwd}}(\cdot)$ represents the forward sequence generation, $\mathcal{N}(\cdot)$ means the layer normalization (LN), $\varphi(\cdot)$ indicates the SiLU activation function, $\mathcal{C}_{\text{onv}}(\cdot)$ is the 1-D convolution, $\mathcal{S}_{SM}(\cdot)$ symbolizes the state space model, $\otimes$ is element-wise multiplication, and $F_{\text{fwd}}^{\text{MS}} \in \mathbb{R}^{128\times64}$ stands for the forward sequence features obtained by forward scanning the input Mamba. Similarly, the reverse $F_{\text{rev}}^{\text{MS}} \in \mathbb{R}^{128\times64}$ and random $F_{\text{ran}}^{\text{MS}} \in \mathbb{R}^{128\times64}$ sequence features can be obtained, respectively. Then, the multipath selective scanning spectral features $F_{\text{spe}}^{m} \in \mathbb{R}^{128\times64}$ are obtained by averaging pooling (AP) the sequences in three different directions. This is mainly because it can retain the average characteristics of these channel sequences, thus more comprehensively reflecting the overall distribution of spectral information

$$F_{\text{spe}}^{m} = (F_{\text{fwd}}^{\text{MS}} + F_{\text{rev}}^{\text{MS}} + F_{\text{ran}}^{\text{MS}})/3. \tag{5}$$

2) Spatial Mamba: Compared with MS images, PAN images contain rich spatial information. Similarly, this article designs a spatial Mamba: a selective scanning structured state space model for spatial sequences. The specific design is shown in Fig. 3(b). Unlike spectral Mamba, the spatial Mamba uses scanning in different directions and dimensions to expand the image patch to four different independent spatial sequences: horizontal forward and reverse, as well as vertical forward and reverse order. Then, they are input into Mamba, and the multipath selective scanning spatial feature $F_{\text{spa}}^{m} \in \mathbb{R}^{128\times64}$ is obtained by average pooling and fully connected layers, which can capture the average distribution of different spatial sequence features and help preserve the details in the image.

In summary, in the feature extraction stage, Fig. 3(a) and (b) is two parallel branches without parameter sharing. The three spectral sequences of forward, reverse, and random in Fig. 3(a); and the four spatial sequences of horizontal forward and reverse, as well as vertical forward and reverse in Fig. 3(b) are, respectively, sent to seven Mamba models [see Fig. 3(c)] with the same structure but no shared parameters. Subsequently, the outputs of the spectral and the spatial dimensions are fused by average pooling to achieve effective information integration.

Regarding image feature learning, the multipath scanning mechanism in Mamba achieves balanced processing of local

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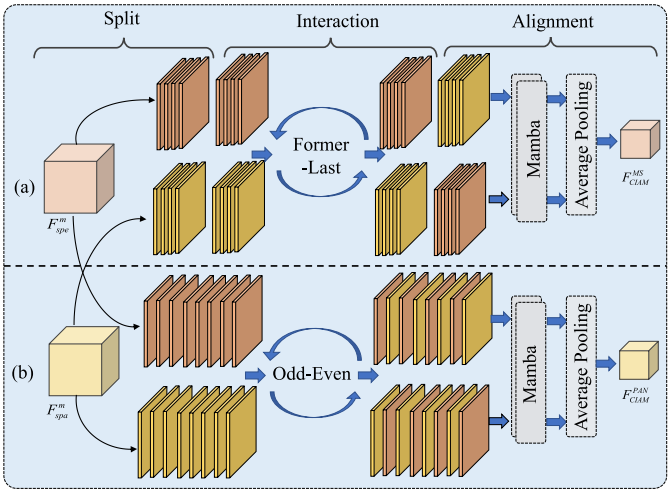


Fig. 4. CIAM. (a) Means former-last interaction. (a) Odd-even interaction.

and global features through convolutional state prediction and selective scanning, while maintaining low computational complexity and good scalability. It is particularly good at processing images without strong sequential relationships. The standard CNN convolution kernel is good at extracting local features and constructing global features through a hierarchical structure, which is suitable for fine feature extraction of ordered data. The Transformer self-attention mechanism directly captures global dependencies. It is suitable for tasks that need to capture long-distance dependencies, but there are challenges in processing local features and computational complexity.

The SCPF [17], MSFN [48], LSPGF [49], DSSN [50], and CDSFT [51] are models based on pure CNN or Transformer to capture spatial and spectral features. However, these methods can usually only capture local or global features with high computational complexity. Therefore, this article proposes the MPS^{2}M, which designs spectral Mamba (selective scanning structured state space model for spectral sequence) and spatial Mamba (selective scanning structured state space model for spatial sequence) for MS and PAN images, respectively. It can avoid the model's excessive dependence on the specific order of channels and patches, thus improving the spectral and spatial characterization capabilities.

C. Channel Interaction Alignment Module

With the rapid development of remote sensing technology, MS and PAN images are being collected from various satellite platforms, carrying unique advantages and details, but also highlighting significant inconsistencies in spatial feature representation. In order to effectively fuse multimodal remote sensing image features, reduce differences between modalities, and improve the alignment of latent spatial representation, this article proposes a CIAM (see Fig. 4). The specific design process is as follows.

1) *Split:* It first divides the channels of F^m_{spe} and F^m_{spa}, respectively. The division strategies include the former-last channel division [see Fig. 4(a)] and odd-even channel division [see Fig. 4(b)] to capture feature information of different dimensions.
2) *Interaction:* Subsequently, these divided channels interact cleverly, allowing the features of different spaces to be shared and fused, as well as the features of different modalities to penetrate and flow with each other, achieving a seamless exchange of cross-modal information and effectively narrowing the inherent differences between modalities.
3) *Alignment:* Next, the four sequences that have been interacted are input into the Mamba module, which can not only deeply model the complex dependencies between channels, but also further reduce the small gaps in spatial representation while maintaining feature diversity, achieving accurate alignment and efficient fusion of features. Finally, the four sequences obtained by inputting the Mamba module are average pooling to obtain features $F^{\text{MS}}_{\text{CIAM}} \in \mathbb{R}^{128\times 64}$ and $F^{\text{PAN}}_{\text{CIAM}} \in \mathbb{R}^{128\times 64}$ of channel interaction alignment.

There are significant differences in the imaging mechanisms of MS and PAN images. At the same time, as the deep network extracts image features layer by layer, the number of channels gradually increases, while the spatial resolution decreases accordingly. As a result, in the deep stage of the network, the primary information is no longer the original spatial pixel value, but rather highly abstracted and encoded channel features. Therefore, this article improves the similarity between MS and PAN images by interactively dividing the former-last and the odd-even channels.

The CIAM can integrate the complementary information of multimodal images through different channel interactions. Thus, it realizes the effective semantic alignment of spatial representation. This provides strong support for subsequent advanced classification tasks.

D. Semantic Aware Calibration Module

With the continuous development of deep feature extraction technology, deep networks are usually able to capture rich semantic information about objects. In order to further optimize these features, especially for the singular or unique features that may appear in the fusion process, this article is inspired by the design of the cross-attention mechanism [52] and innovatively proposes a SACM (as shown in Fig. 5). The SACM module aims to accurately calibrate singular or unique features through interactive enhancement with shared features, thereby ensuring that the loss of complementary information is minimized while retaining their respective advantageous attributes. The specific design process is as follows.

This article takes the singular features $F^{\text{MS}}_{\text{CIAM}}$ and the fused features F^{Fu}_{CIAM} as examples. First, these features are passed through the CIAM, followed by average pooling (fusion of CIAM outputs), LN, and MLP. Concurrently, F^{Fu}_{CIAM} undergoes LN and MLP to map them within a shared space

$$\begin{aligned} x, z &= \mathcal{M}_{lp}(\mathcal{N}(\mathcal{A}p(f_{\text{CIAM}}(F^{\text{MS}}_{\text{CIAM}}, F^{Fu}_{\text{CIAM}})))) \\ y &= \mathcal{M}_{lp}(\mathcal{N}(F^{Fu}_{\text{CIAM}})). \end{aligned} \tag{6}$$

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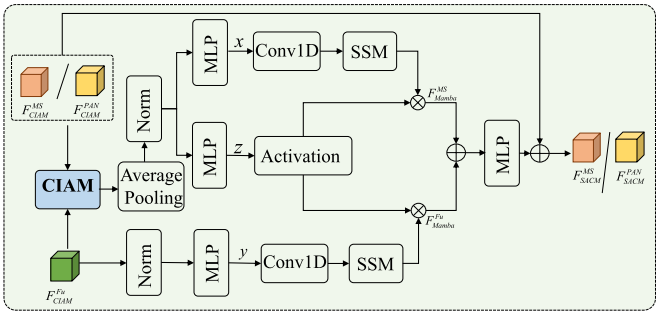


Fig. 5. SACM.

Algorithm 1 SACM

Require: token sequence $F_{\text{CIAM}}^{\text{MS}}$, $F_{\text{CIAM}}^{\text{PAN}}$, F_{CIAM}^{Fu}: (N, C)
Ensure: token sequence $F_{\text{SACM}}^{\text{MS}}$, $F_{\text{SACM}}^{\text{PAN}}$: (N, C)

```
MambaIndex = 0
for MambaIndex < MambaCount do
    /* CIAM, AP, layer normalization and MLP */
    x, y = M_lp(N(Ap(f_CIAM(F^MS_CIAM, F^Fu_CIAM))))
    z = M_lp(N(F^Fu_CIAM))
    /* Process with sequence */
    F^MS_Mamba = φ(z) ⊗ S_sm(C_onv(x))
    F^Fu_Mamba = φ(z) ⊗ S_sm(C_onv(y))
    /* Residual connection*/
    F^MS_SACM = M_lp(F^MS_Mamba + F^Fu_Mamba) + F^MS_CIAM
    MambaIndex = MambaIndex + 1
end for
/* The same principle can be obtained*/
for MambaIndex < MambaCount do
    x', y' = M_lp(N(Ap(f_CIAM(F^PAN_CIAM, F^Fu_CIAM))))
    z' = M_lp(N(F^Fu_CIAM))
    F^PAN_Mamba = φ(z') ⊗ S_sm(C_onv(x'))
    F^Fu_Mamba = φ(z') ⊗ S_sm(C_onv(y'))
    F^PAN_SACM = M_lp(F^PAN_Mamba + F^Fu_Mamba) + F^PAN_CIAM
    MambaIndex = MambaIndex + 1
end for
```

where $f_{\text{CIAM}}(\cdot)$ represents the CIAM block. $\mathcal{A}p(\cdot)$ is the average pooling.

Then, x and y are input into different 1-D convolutions and SSM. In addition, z is processed through the SiLU activation function, which provides convenience for the calibration of fusion features to singular complementary features

$$F_{\text{Mamba}}^{\text{MS}} = \varphi(z) \otimes \mathcal{S}_{sm}(\mathcal{C}_{\text{onv}}(x))$$
$$F_{\text{Mamba}}^{Fu} = \varphi(z) \otimes \mathcal{S}_{sm}(\mathcal{C}_{\text{onv}}(y)). \tag{7}$$

Finally, $F_{\text{Mamba}}^{\text{MS}} \in \mathbb{R}^{128\times64}$ and $F_{\text{Mamba}}^{Fu} \in \mathbb{R}^{128\times64}$ are added and input into the MLP, as well as then added to the original $F_{\text{CIAM}}^{\text{MS}}$ to obtain the fused calibrated feature $F_{\text{SACM}}^{\text{MS}} \in \mathbb{R}^{128\times64}$

$$F_{\text{SACM}}^{\text{MS}} = \mathcal{M}_{lp}(F_{\text{Mamba}}^{\text{MS}} + F_{\text{Mamba}}^{Fu}) + F_{\text{CIAM}}^{\text{MS}}. \tag{8}$$

Similarly, inputting $F_{\text{CIAM}}^{\text{PAN}}$ and F_{CIAM}^{Fu} into SACM module can also get $F_{\text{SACM}}^{\text{PAN}} \in \mathbb{R}^{128\times64}$.

Compared with the cross-attention mechanism based on self-attention [52], [53], [54], the SACM (see Algorithm 1) aims to interact fusion features and the z-vectors of the unique features in the Mamba architecture, significantly reduces the computational complexity while retaining accuracy through refined algorithm design, thereby improving processing efficiency. In addition, the module achieves accurate calibration of unique features by efficiently fusing multisource features. This process not only retains the advantageous information in the features of each modality, but also significantly enhances the complementary information between them that may have been overlooked, effectively preventing the loss of complementary content in the information transmission of a single modality. Consequently, the SACM can promote more comprehensive and integrated feature extraction, which is more conducive to the fusion classification of multimodal remote sensing images.

IV. Experiments

This chapter mainly introduces the experimental settings, parameter selection, ablation experiments, and comparisons with other methods in detail.

A. Experimental Settings

1) Datasets: This article uses three datasets (see Fig. 6) to verify the effectiveness of the proposed method: Xi'an Urban, Xi'an, and Nanjing datasets.

1) *Xi'an urban [see Fig. 6(a)]:* The MS image is characterized by dimensions of 800 × 830 pixels, encompassing four spectral bands with an 8m resolution. Conversely, the PAN image exhibits dimensions of 2400 × 2520 pixels and consists of a single-channel spectral band with a finer 2-m resolution. Within the dataset, labels have been provided for seven distinct types of ground scenes, totaling 294 468 samples. Notably, 7774 samples are earmarked for training purposes, leaving a substantial 286 694 samples for subsequent testing and evaluation.
2) *Xi'an [see Fig. 6(b)]:* The MS image measures 4548 × 4541×4 pixels and has four spectral bands with an 8 m resolution. In contrast, the PAN image has dimensions of 18 192 × 18 164 × 1 pixels and comprises a single-channel spectral band with a sharper 2-m resolution. The collection includes labels for 12 different categories of ground scenes, totaling 2 464 529 samples. Notably, 55 892 samples are set aside for training, leaving a significant 2 408 637 samples for later testing and evaluation.
3) *Nanjing [see Fig. 6(c)]:* The dataset provides the MS image measuring 2000 × 2500 pixels and containing four spectral bands at an 8m resolution. In comparison, there is the PAN image with dimensions of 8000 × 10 000 pixels and a single-channel spectral band with a better 2-m resolution. The dataset contains labels representing eleven different ground scene classes, for a total of 476 879 samples. It is worth noting that 14 018 samples are set aside for training, while the remaining 462 861 samples are kept for future testing and evaluation.

2) Training Parameter Settings: During the experimental phase, specific parameters are configured for the study. Notably, the initial learning rate is initialized at 0.0002 to guide

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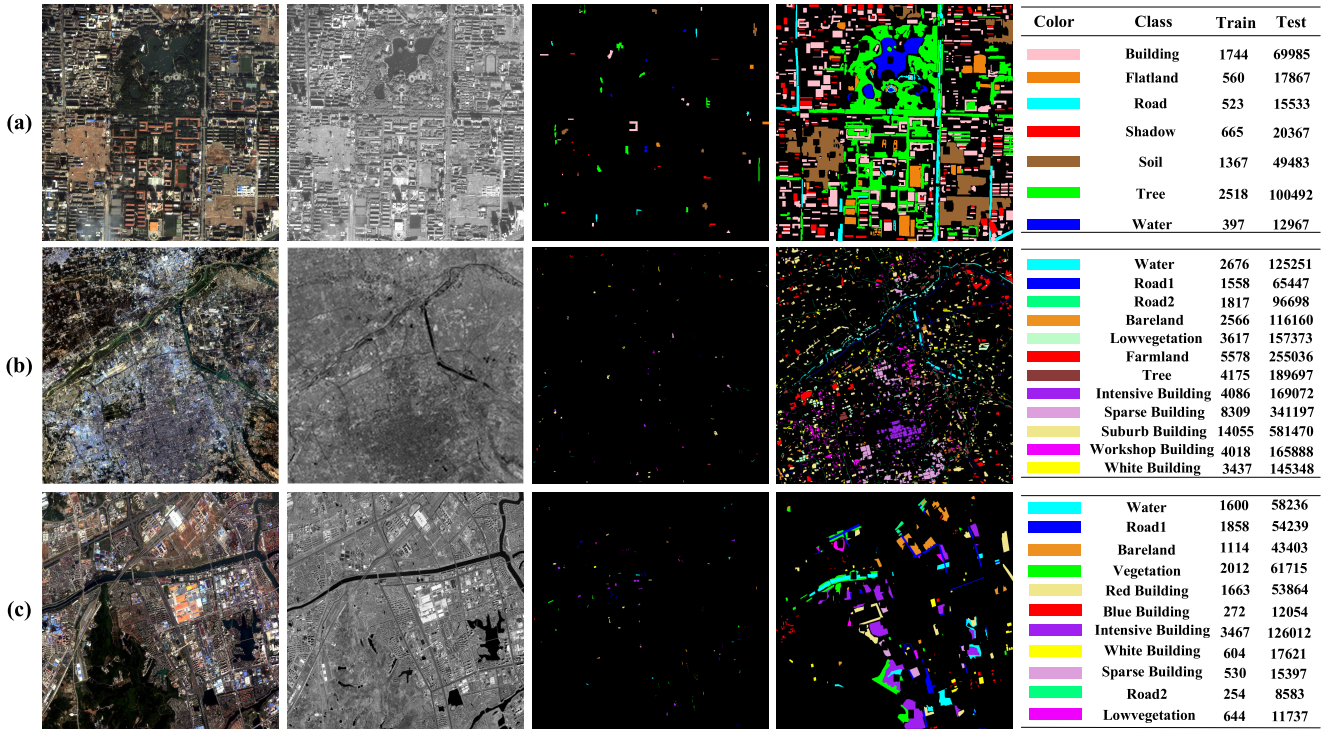


Fig. 6. Visualization of datasets. From top to bottom are the scenes of Xi'an Urban, Xi'an, and Nanjing. From left to right are MS images, PAN images, training sets, test sets, and related data statistics.

the training process, and the weight decay is set to 0.001, with a total of 40 epochs established for this stage. In the Xi'an Urban and Nanjing datasets, a batch size of 128 is employed, while a batch size of 256 is utilized for the Xi'an dataset. The optimization function leverages the adaptive moment estimate (Adam). The experimental investigations are conducted on an AMAX workstation operating Ubuntu 16.04 LTS, equipped with an NVIDIA 2080Ti GPU to expedite computations. The PyTorch framework, in conjunction with Python 3.9, serves as the primary tool for executing the diverse experimental tasks within this research framework.

3) Quality Evaluation Indicators: Throughout the research process, we will systematically evaluate multiple key performance indicators to comprehensively measure the effectiveness and reliability of the model. Specifically, we will focus on the overall accuracy (OA), which measures the overall correctness of the model's prediction results for all categories; at the same time, by calculating the average accuracy (AA), we can evaluate the average prediction ability of the model in each category, thereby identifying potential category bias problems; in addition, the Kappa coefficient statistic (Kappa) will be introduced, which takes into account the impact of random consistency on accuracy and provides a more robust perspective for model performance evaluation. The definitions for these metrics are as follows.

1) *OA:*

$$\mathrm{OA} = \frac{\sum_{i=1}^{Y} q_{ii}}{Q} \tag{9}$$

where Y refers to the total number of different categories in the entire dataset. q_{ii} is used to record the number of samples belonging to the ith category and accurately predicted to be of the same category. Q is the total number of samples in all categories.

2) *AA:*

$$\mathrm{AA} = \frac{1}{Y} \sum_{i=1}^{Y} \frac{n_{ii}}{n_i} \tag{10}$$

where n_{ii} is the number of samples that are accurately identified and classified in the ith category. n_i represents the total number of all samples contained in the ith category.

3) *Kappa:*

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{PE}}{1 - \mathrm{PE}} \tag{11}$$

where PE is the expected consistency ratio.

B. Parameter Determination

Within this chapter's exploration of network module number influence on overall performance, the focus is directed toward optimizing various module combination strategies to enhance network efficiency through meticulous parameter tuning. The investigation specifically delves into the amalgamation of different modules, varying their numbers to determine the optimal configuration for maximizing network performance. The selection of the numbers N_1, N_2, N_3, and N_4 of spectral Mamba, spatial Mamba, CIAM, and SACM is facilitated by the control variable method. Through rigorous experimentation across three datasets (see Table I), irrespective of the dataset under scrutiny, the most favorable results consistently emerge

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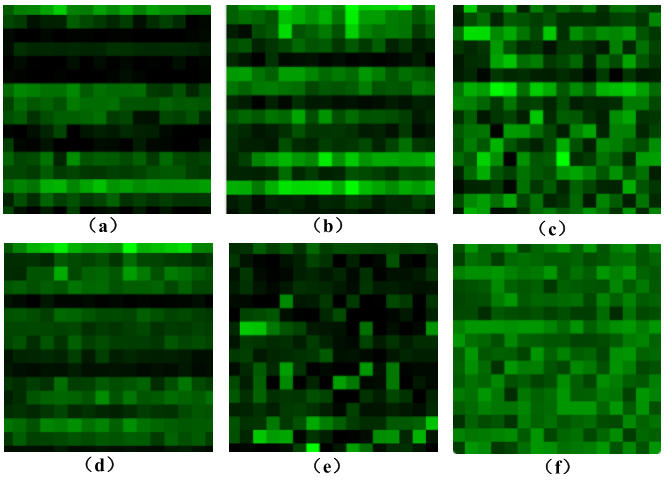


Fig. 7. Visualization results of ablation experimental on Xi'an Urban. (a) and (b) Visualization feature maps of MS images before and after inputting Spectral Mamba. (c) Visualization feature map after CIAM. (d) and (e) Visualization feature maps of PAN images before and after inputting Spatial Mamba. (f) Visualization feature map after SACM.

when the module configuration is set to [2, 2, 2, 1]. This validation underscores the significance of specific module composition in attaining superior performance across varied datasets.

C. Ablation Experiment

In this study, we have randomly chosen the Xi'an Urban dataset for a comprehensive combinatorial investigation involving the MPS^{2}M: spectral Mamba and spatial Mamba, the CIAM, as well as the SACM. By conducting experiments with various combinations of these modules, we aim to assess their individual effectiveness and necessity within the network architecture. The detailed results of these experiments, along with comparative analyses of different module configurations, are meticulously documented in Table II, providing insights into the impact and significance of each module in optimizing network performance on the Xi'an Urban dataset.

In the experimental verification on the Xi'an Urban dataset, the CMS^{2}I framework achieves the best performance (OA: 84.04%, AA: 77.81%, and Kappa: 79.43%) through the close collaboration of its five core modules (Mamba ∪ Spectral Mamba ∪ Spatial Mamba ∪ CIAM ∪ SACM). This is because MPS^{2}M can use spectral Mamba and spatial Mamba to simultaneously model global dependencies with scanning mechanisms of different dimensions for the MS and the PAN image characteristics. The CIAM can effectively reduce the differences between multimodal data through the former-last and odd-even channel interaction strategies, thereby achieving multimodal spatial semantic alignment as much as possible. In addition, the SACM can use fused features to calibrate two singular features, avoiding further expansion of modal space differences. This comprehensive solution works together on CMS^{2}I-Mamba, enabling it to show excellent performance in processing multisource remote sensing image fusion classification.

Fig. 7 depicts the feature map visualization of Xi'an Urban dataset with different network layers. As we all know, the lighter the hue, the stronger the spectral intensity, and the higher the attention. It is obvious that as the network is trained, the spectral intensity becomes more concentrated, and distinct features as well as good spatial allocations are prioritized, demonstrating the superiority of the spectral Mamba, spatial Mamba, CIAM, and SACM in feature extraction. Meanwhile, the texture information of the features is significantly increased after passing through these modules.

D. Comparison With Other Methods on Different Datasets

The proposed CMS^{2}I-Mamba is compared with seven state-of-the-art algorithms CRHFF [41], AM^{3}Net [42], GCFNet [43], MFT [44], DFINet [45], ISSP [55], and NNCNet [46]. All quantitative results presented are the average of five experiments. The experimental results of different scenarios are as follows.

1) Experimental Results in Xi'an Urban: Table III and Fig. 8 show the indicator calculation and classification visualization on the Xi'an Urban dataset, respectively.

The CRHFF [41] aims to extract and fuse cross-resolution latent information, thereby enhancing the modeling of imaged objects from the multimodal representation and ultimately increasing classification accuracy, by integrating both shallow-to-deep and global-to-local features from high-resolution MS and PAN images. It achieves higher accuracy on c_5: soil (90.99%) and c_7: water (98.04%) categories.

The AM^{3}Net [42] can independently measure the contribution of each pixel's spectral channel to the spectral feature tensor, promote feature co-transfer between spectral features and spatial information, as well as enhance semantic relevance through adaptive, multiscale, and mutual learning processes. Compared with CRHFF, it has made progress in c_1: building (84.27% versus 77.53%) and c_6: tree (82.43% versus 79.98%).

The GCFnet [43] is an encoder–decoder-based deep learning network designed for global patchless image classification and can capture image context. In complex urban dataset tests, it improves performance by fusing sampled and unsampled location data. Compared with AM^{3}Net, it has achieved an order of magnitude improvement on c_5: soil (80.74% versus 55.32%).

The MFT [44] adopts the mCrossPA mechanism to combine HSI with other complementary information through tokenization technology to learn better representation in a simplified feature space, thereby improving the HSI land cover classification performance. Although it does not have the highest accuracy category, the OA is relatively average, and there is no worst category.

The DFINet [45] fuses multisource remote sensing data through a deep cross-attention module, extracts meaningful autocorrelation and cross correlation information, as well as achieves improved classification accuracy by coordinating consistency, discrimination, and classification loss optimization. Therefore, it achieves the highest category accuracy on c_2: flatland (52.87%), c_3: road (80.65%), c_5: soil (92.98%), and c_7: water (98.71%).

The NNCNet [46] uses unlabeled data through self-supervised learning to overcome the problem of inconsistent distribution of multisource data. It uses nearest

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TABLE I

EXPERIMENTAL RESULTS OF THE CMS^{2}I-MAMBA HYPERPARAMETER BLOCK NUMBERS N_1, N_2, N_3, N_4

Dataset	Xi'an Urban			Xi'an			Nanjing		
N_1, N_2, N_3, N_4	[1,1,2,1]	[2,2,2,1]	[2,2,2,2]	[1,1,2,1]	[2,2,2,1]	[2,2,2,2]	[1,1,2,1]	[2,2,2,1]	[2,2,2,2]
OA (%)	83.42	**84.04**	83.29	80.03	**80.20**	79.92	79.58	**80.07**	79.63
AA (%)	77.18	**77.81**	77.34	73.65	**73.83**	73.29	75.89	**76.65**	76.27
Kappa (%)	79.25	**79.43**	79.16	77.02	**77.32**	76.93	75.90	**76.76**	76.53

TABLE II

RESULT OF THE ABLATION STUDY ON THE XIAN URBAN IMAGE

	Mamba	Spectral Mamba	Spatial Mamba	CIAM	SACM	OA (%)	AA (%)	Kappa (%)
(1)	√					81.09	75.73	76.89
(2)	√	√				82.59	76.92	77.58
(3)	√		√			82.62	76.91	77.76
(4)	√	√	√			83.29	77.16	78.23
(5)	√	√	√	√		83.57	77.46	78.95
(6)	√	√	√	√	√	**84.04**	**77.81**	**79.43**

TABLE III

QUANTITATIVE RESULTS OF THE XI'AN URBAN DATASET

Models	CRHFF [41]	AM^{3}Net [42]	GCFNet [43]	MFT [44]	DFINet [45]	NNCNet [46]	ISSP [55]	CMS^{2}I-Mamba
c_1: Building	77.53	84.27	72.96	74.96	71.58	**85.62**	76.39	85.37
c_2: Flatland	40.93	35.89	30.75	46.82	**52.87**	38.35	37.21	38.67
c_3: Road	73.86	39.09	25.46	43.74	**80.65**	59.89	64.04	60.15
c_4: Shadow	75.48	58.12	42.02	68.17	45.33	71.41	**83.35**	80.86
c_5: Soil	90.99	55.32	80.74	81.70	**92.98**	90.42	89.72	91.84
c_6: Tree	79.98	82.43	88.36	82.21	85.55	**92.55**	83.22	89.90
c_7: Water	98.04	96.86	71.37	98.09	**98.71**	95.61	97.89	97.91
OA(%)	79.01	71.87	72.23	75.78	78.86	83.98	79.44	**84.04**
AA(%)	76.69	64.57	58.81	70.81	75.38	76.26	75.97	**77.81**
Kappa(%)	73.35	63.19	63.97	68.89	72.82	79.31	73.80	**79.43**
Time(s)	127.99	176.56	268.79	**17.63**	49.72	88.15	238.15	52.00
Parameter(n)	4906823	2777838	3878187	**212876**	759438	2212423	12082537	1931527

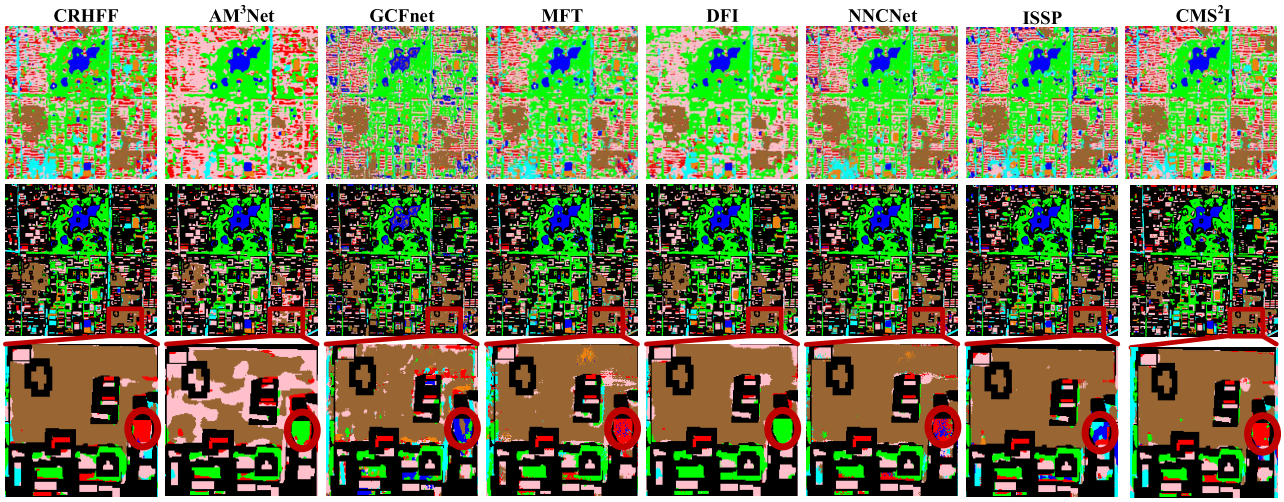


Fig. 8. Visualization results of experimental on Xi'an Urban. Each column contains an overall color map and a label position color map. From left to right, they are CRHFF [41], AM^{3}Net [42], GCFNet [43], MFT [44], DFINet [45], NNCNet [46], ISSP [55], CMS^{2}I-Mamba. The middle area of the close-up image is composed of a part of "c_4: Shadow," "c_6: Tree," and "c_7: Water."

neighbor data enhancement and bilinear attention modules to enhance semantic similarity and feature interaction, respectively, improving classification results. It achieves the highest category accuracy on c_1: building (85.62%) and c_6: tree (92.55%) categories.

The ISSP [55] uses a pixel-guided spatial enhancement module to improve spatial position interaction, fuse multi-scale contextual information, and highlight pixel-level features. At the same time, it uses a time–frequency collaborative spectral enhancement module to utilize the main and auxiliary

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TABLE IV
QUANTITATIVE RESULTS OF THE XI'AN DATASET

Models	CRHFF [41]	AM^{3}Net [42]	GCFNet [43]	MFT [44]	DFINet [45]	NNCNet [46]	ISSP [55]	CMS^{2}I-Mamba
c_1: Water	95.89	95.61	95.89	**97.35**	93.51	93.15	96.75	96.14
c_2: Road1	18.35	1.72	17.46	16.82	15.41	**34.16**	26.29	28.21
c_3: Road2	62.01	11.61	55.38	50.69	53.66	62.14	**68.75**	54.50
c_4: Bareland	66.45	67.74	66.46	64.58	75.55	69.01	**72.70**	68.46
c_5: Lowvegetation	**55.51**	18.81	36.06	46.68	45.00	55.10	54.81	51.11
c_6: Farmland	69.74	**76.35**	64.31	51.69	50.59	68.41	71.42	73.85
c_7: Tree	84.84	37.73	89.12	91.78	**92.29**	77.92	86.47	89.48
c_8: Intensive Building	77.59	72.44	72.65	62.85	**86.75**	74.62	74.57	81.19
c_9: Sparse Building	77.18	**80.85**	76.40	75.73	78.26	77.48	77.17	76.64
c_{10}: Suburb Building	95.51	**99.81**	97.81	91.90	96.06	96.81	95.47	95.65
c_{11}: Workshop Building	95.39	75.48	90.71	94.52	**96.75**	94.23	92.03	93.03
c_{12}: White Building	75.33	45.00	66.36	72.87	**83.25**	76.02	67.29	77.63
OA (%)	79.42	69.58	76.86	74.65	78.70	79.24	79.58	**80.20**
AA (%)	72.81	56.93	69.05	68.12	72.26	73.25	73.64	**73.83**
Kappa (%)	76.44	64.18	73.48	71.08	75.78	76.24	76.66	**77.32**
Time(s)	1159.44	1041.57	1159.44	**140.23**	425.58	756.16	872.85	602.27

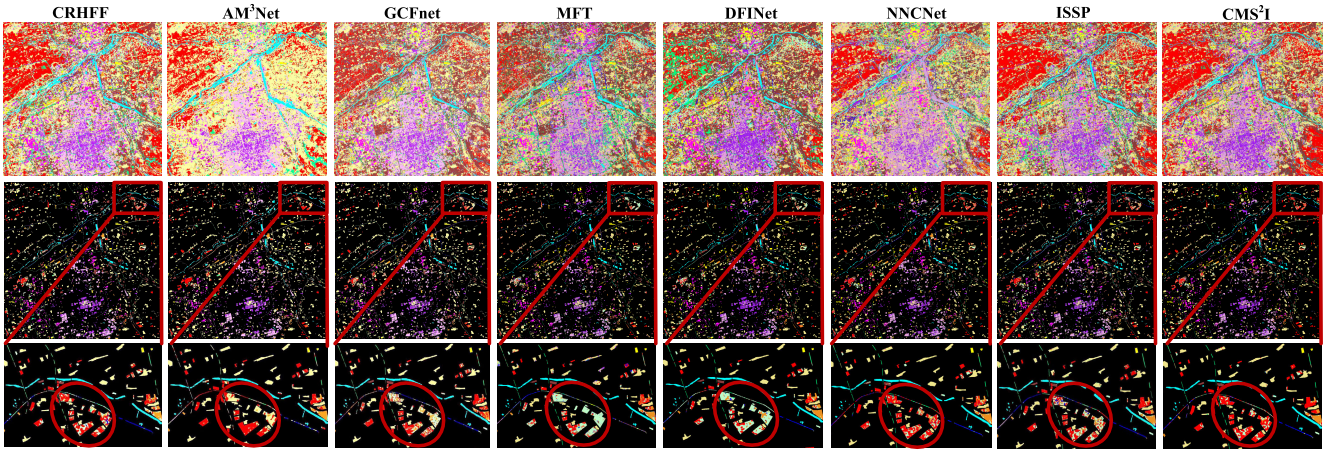


Fig. 9. Visualization results of experimental on Xi'an. Each column contains an overall color map and a label position color map. From left to right, they are CRHFF [41], AM^{3}Net [42], GCFNet [43], MFT [44], DFINet [45], NNCNet [46], ISSP [55], CMS^{2}I-Mamba. The middle area of the close-up image is composed of a part of "c_6: Farmland" and "c_{10}: Suburb building."

information of multimodal data comprehensively. Therefore, it achieves the highest classification accuracy in category c_4: shadow (83.35%).

The CMS^{2}I-Mamba captures global long-range dependencies with linear complexity. Among them, the MPS^{2}M adopts flexible and changeable spatial and spectral scanning strategies to strengthen the information related to the classification task. The CIAM adopts a variety of channel interaction mechanisms to promote the flow and fusion of information between different channels, which is conducive to the alignment of different modal spaces. In addition, the semantic awareness calibration module (SACM), inspired by cross-attention, uses fused features to calibrate singular features, effectively preventing the loss of complementary content in a single modality during information transmission. Therefore, CMS^{2}I-Mamba can clearly identify c_4 from c_4: shadow, c_6: tree, and c_7: water based on the texture information. Since different objects occupy different proportions in the image, CMS^{2}I-Mamba does not achieve the highest category accuracy in some specific categories. However, CMS^{2}I-Mamba aims to comprehensively balance the representation capabilities of features from both spatial and spectral perspectives, providing a more robust and comprehensive solution, which makes it better than those models that only focus on a single feature extraction method in terms of comprehensive performance. That is, CMS^{2}I-Mamba's overall OA (84.04%), AA (77.81%), and Kappa (79.43%) are the highest.

2) Experimental Results in Xi'an: Table IV and Fig. 9 show the indicator calculation and classification visualization on the Xi'an dataset, respectively.

By comparing the performance of different models, it can be seen that CMS^{2}I-Mamba performs best in overall performance indicators, with OA, AA, and Kappa of 80.20%, 73.83%, and 77.32%, respectively, all higher than other models. Specifically for each category, the performance of each model in different categories is different. For example, MFT excels in c_1 (water: 97.35%) category, while NNCNet demonstrates superior performance in c_2 (road1: 34.16%) categories. In other categories, such as c_5 (lowvegetation: 55.51%), CRHFF performs best accuracy, while AM^{3}Net achieves the highest accuracy in c_6 (farmland: 76.35%), c_9 (sparse building: 80.85%), and c_{10} (suburb building: 99.81%). The DFINet performs best in c_4 (bareland: 75.55%), c_7 (tree: 92.29%), c_8 (intensive building: 86.75%), c_{11} (workshop building: 96.75%), and c_{12} (white

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TABLE V
QUANTITATIVE RESULTS OF THE NANJING DATASET

Models	CRHFF [41]	AM^{3}Net [42]	GCFNet [43]	MFT [44]	DFINet [45]	NNCNet [46]	ISSP [55]	CMS^{2}I-Mamba
c_1: Water	94.86	88.44	93.45	94.23	**96.18**	94.77	94.24	95.27
c_2: Road1	**72.96**	40.62	37.36	64.21	72.09	67.25	59.47	70.11
c_3: Bareland	94.24	54.10	84.98	91.84	92.23	88.19	**94.50**	92.72
c_4: Vegetation	84.85	71.55	78.65	76.58	**88.77**	75.83	80.80	88.25
c_5: Red Building	77.61	67.85	24.98	75.54	85.50	77.94	78.04	**88.04**
c_6: Blue Building	95.59	93.66	44.95	91.60	97.09	93.03	95.33	**97.89**
c_7: Intensive Building	59.08	74.96	64.42	64.48	64.35	**75.72**	69.34	70.90
c_8: White Building	81.76	90.98	64.74	75.73	86.71	84.37	58.44	**91.88**
c_9: Sparse Building	53.61	**80.99**	45.62	38.92	44.81	47.68	52.80	47.07
c_{10}: Road2	43.07	**91.86**	13.58	34.21	30.92	53.03	33.90	45.84
c_{11}: Lowvegetation	62.74	43.96	47.84	63.27	63.27	**66.78**	59.09	55.15
OA(%)	75.52	70.22	61.65	73.35	78.00	77.77	75.01	**80.07**
AA(%)	74.85	72.63	54.60	70.06	74.72	74.96	70.54	**76.65**
Kappa(%)	71.77	65.02	55.01	69.11	74.47	74.03	71.03	**76.76**
Time(s)	206.27	140.97	198.96	**28.57**	79.11	142.89	286.71	70.93

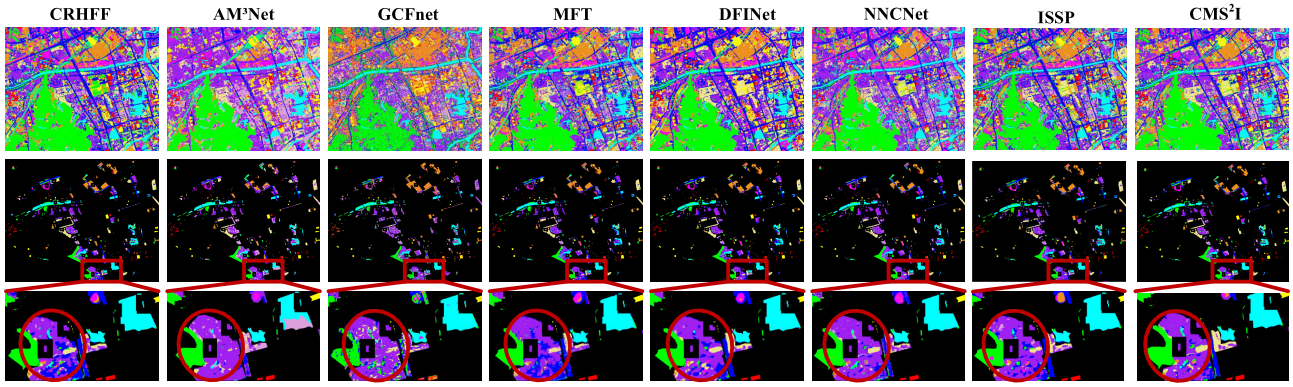


Fig. 10. Visualization results of experimental on Nanjing. Each column contains an overall color map and a label position color map. From left to right, they are CRHFF [41], AM^{3}Net [42], GCFNet [43], MFT [44], DFINet [45], NNCNet [46], ISSP [55], CMS^{2}I-Mamba. The middle area of the close-up image is composed of a part of "c_1: Water," "c_2: Road1," "c_5: Red Building," and "c_7: Intensive Building."

building: 83.25%). It is worth noting that ISSP achieves the highest accuracy in c_3 (road2: 68.75%). Simultaneously, it can be seen from the visualization of Fig. 9 that the proposed method can effectively distinguish between "farmland" and "suburb building" with similar appearance. Nevertheless, CMS^{2}I-Mamba also performs well in multiple categories, especially ranking high in categories such as c_1 (water), c_8 (intensive building), c_9 (sparse building), and c_{10} (suburb building). In summary, although CMS^{2}I-Mamba does not have the highest category accuracy, it is still leading in overall performance.

3) Experimental Results in Nanjing: Table V and Fig. 10 show the indicator calculation and classification visualization on the Nanjing dataset, respectively.

The CMS^{2}I-Mamba leverages exceptional feature interaction capabilities, leading to optimal classification accuracy in over one-fourth of the categories. Notably, our proposed CMS^{2}I-Mamba demonstrates superior performance with an OA of 80.07%, AA of 76.65%, and Kappa coefficient of 76.76%. Compared to NNCNet, CMS^{2}I-Mamba showcases improvements of 2.3%, 1.69%, and 2.73% in OA, AA, and Kappa, respectively. The CRHFF utilizes an encoder to extract multichannel and high-resolution features, excelling in global background feature extraction and yielding impressive results in categories c_2 (road1, 72.96%). The spatial location capturing ability of AM^{3}Net contributes to excellent categorization in c_9 (sparse building, 80.99%) and c_{10} (road2, 91.86%). The DFINet exhibits spectral collection capabilities, reflecting superior performance in c_1 (water, 96.18%) and c_4 (vegetation, 88.77%). The NNCNet enhances neighborhood information using a bilinear attention module, achieving peak accuracy in categories c_7 (intensive building, 75.72%) and c_{11} (low vegetation, 66.78%). The ISSP has the highest precision in c_3 (bareland, 94.50%). The CMS^{2}I-Mamba exhibits the highest classification accuracy in c_5 (red building, 88.04%), c_6 (blue building, 97.89%), and c_8 (white building, 91.88%). According to the visualization results, CMS^{2}I-Mamba can identify scene objects (c_1, c_2, c_5, and c_7) from multiple dimensions such as color saturation, density, and shape. This is mainly due to the fact that CMS^{2}I-Mamba can achieve the recognition of spectral details through different channel interaction strategies. These results further prove that the overall classification performance of the CMS^{2}I-Mamba on the Nanjing dataset is better than other models, and has better generalization ability and robustness.

Based on a large amount of experimental data analysis, the accuracy achieved by CMS^{2}I-Mamba significantly exceeds that of most comparable algorithms. In the tests

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of all datasets, it achieves the highest indicator calculation. Meanwhile, the CMS^{2}I-Mamba shows high efficiency, especially in the test phase. Although it does not have the least number of parameters, its test time performance on multiple datasets is good, indicating that CMS^{2}I-Mamba maintains good computational efficiency while achieving high accuracy. Therefore, CMS^{2}I-Mamba can also complete classification tasks at a faster speed in practical applications, which is particularly important for the real-time interpretation of multimodal images.

V. Conclusion

This article proposes a CMS^{2}I-Mamba for the fusion classification of multimodal remote sensing images, which can improve the representation ability of features through interaction, alignment, and calibration, thereby improving classification accuracy. The MPS^{2}M can adapt to texture and edge changes in actual scenes with variable spectral and spatial scanning strategies. The CIAM achieves effective alignment of spatial representations through former-last and odd-even channel interactive ways. The SACM aims to guide shared features to calibrate singular features, thereby reducing information loss. Through verification on three different scene datasets, CMS^{2}I-Mamba achieves impressive results compared with other comparison algorithms. In the near future, we will try to retain the current accuracy as much as possible while improving the generalization performance of the model.

References

[1] J. Long et al., "Joint spatial–spectral optimization for the high-magnification fusion of hyperspectral and multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520919.

[2] X. Geng et al., "Multisource joint representation learning fusion classification for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4406414.

[3] Q. Zhu, Z. Li, T. Song, L. Yao, Q. Guan, and L. Zhang, "Unrestricted region and scale: Deep self-supervised building mapping framework across different cities from five continents," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 344–367, Mar. 2024.

[4] L. Thomas Ramos and A. D. Sappa, "Multispectral semantic segmentation for land cover classification: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 14295–14336, 2024.

[5] M. Ma et al., "MBSI-net: Multimodal balanced self-learning interaction network for image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3819–3833, May 2024.

[6] G. Xie, R. Nie, J. Cao, H. Li, and J. Li, "A deep multiresolution representation framework for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517216.

[7] L. Jiao et al., "Brain-inspired remote sensing foundation models and open problems: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 10084–10120, 2023.

[8] M. A. Fadhel et al., "Comprehensive systematic review of information fusion methods in smart cities and urban environments," *Inf. Fusion*, vol. 107, Jul. 2024, Art. no. 102317.

[9] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605111.

[10] L. Jiao et al., "Brain-inspired learning, perception, and cognition: A comprehensive review," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–21, 2024, doi: 10.1109/TNNLS.2024.3401711.

[11] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[12] J. Zhu, Y. Guo, G. Sun, L. Yang, M. Deng, and J. Chen, "Unsupervised domain adaptation semantic segmentation of high-resolution remote sensing imagery with invariant domain-level prototype memory," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603518.

[13] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[14] J. Zhao et al., "Knowledge guided evolutionary transformer for remote sensing scene classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 10368–10384, Oct. 2024.

[15] L. Jiao et al., "Transformer meets remote sensing video detection and tracking: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1–45, 2023.

[16] Y. Shao et al., "CoT: Contourlet transformer for hierarchical semantic segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 132–146, Jan. 2025.

[17] H. Zhu et al., "A spatial-channel progressive fusion ResNet for remote sensing classification," *Inf. Fusion*, vol. 70, pp. 72–87, Jun. 2021.

[18] H. Zhu et al., "A semantically nonredundant continuous-scale feature network for panchromatic and multispectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5407815.

[19] H. Zhu et al., "ConvGRU-based multiscale frequency fusion network for PAN-MS joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5406415.

[20] W. Ma et al., "Intra- and intersource interactive representation learning network for remote sensing images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5401515.

[21] X. Qi et al., "MLRSNet: A multi-label high spatial resolution remote sensing dataset for semantic scene understanding," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 337–350, Nov. 2020.

[22] A. Yang, M. Li, Y. Ding, D. Hong, Y. Lv, and Y. He, "GTFN: GCN and transformer fusion network with spatial–spectral features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 6600115.

[23] L. Xu et al., "A dual-stream transformer with diff-attention for multispectral and panchromatic classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5409114.

[24] M. Ma et al., "A multimodal hyper-fusion transformer for remote sensing image classification," *Inf. Fusion*, vol. 96, pp. 66–79, Aug. 2023.

[25] D. Zhang, Y. Yang, X. Liu, W. Ma, and L. Jiao, "Dense cross-scale transformer with channel learning for remote sensing scene classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 6101–6104.

[26] Z. Chang, Z. Feng, S. Yang, and Q. Gao, "AFT: Adaptive fusion transformer for visible and infrared images," *IEEE Trans. Image Process.*, vol. 32, pp. 2077–2092, 2023.

[27] E. W. Steele, D. R. Reisingy, J. D. Boyd, and A. M. Murphy, "Deep convolutional neural network for multiple smart grid event classification through sliding windows," in *Proc. Int. Conf. Smart Grid Synchronized Meas. Anal. (SGSMA)*, May 2024, pp. 1–6.

[28] C. Li, B. Rasti, X. Tang, P. Duan, J. Li, and Y. Peng, "Channel-layer-oriented lightweight spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504214.

[29] K. Cho et al., "Learning phrase representations using RNN encoder–decoder for statistical machine translation," 2014, *arXiv:1406.1078*.

[30] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[31] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[32] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[33] X. Xie et al., "FusionMamba: Dynamic feature enhancement for multimodal image fusion with mamba," 2024, *arXiv:2404.09498*.

[34] W. Dong et al., "Fusion-mamba for cross-modality object detection," 2024, *arXiv:2404.09146*.

[35] Z. Li, H. Pan, K. Zhang, Y. Wang, and F. Yu, "MambaDFuse: A mamba-based dual-phase model for multi-modality image fusion," 2024, *arXiv:2404.08406*.

[36] J. Wang, M. Zhang, W. Li, and R. Tao, "A multistage information complementary fusion network based on flexible-mixup for HSI-X image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17189–17201, Dec. 2024.

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[37] J. He, Q. Yuan, J. Li, Y. Xiao, and L. Zhang, "A self-supervised remote sensing image fusion framework with dual-stage self-learning and spectral super-resolution injection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 131–144, Oct. 2023.
[38] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.
[39] Y. Gao, M. Zhang, J. Wang, and W. Li, "Cross-scale mixing attention for multisource remote sensing data fusion and classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507815.
[40] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.
[41] S. Liu, H. Zhao, Q. Du, L. Bruzzone, A. Samat, and X. Tong, "Novel cross-resolution feature-level fusion for joint classification of multispectral and panchromatic remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619314.
[42] J. Wang, J. Li, Y. Shi, J. Lai, and X. Tan, "AM^{3}net: Adaptive mutual-learning-based multimodal data fusion network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5411–5426, Aug. 2022.
[43] H. Zhao et al., "GCFnet: Global collaborative fusion network for multispectral and panchromatic image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5632814.
[44] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.
[45] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.
[46] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.
[47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[48] Y. Wu, R. Dian, and S. Li, "Multistage spatial–spectral fusion network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–11, 2024, doi: 10.1109/TNNLS.2024.3460190.
[49] U. A. Bhatti et al., "Local similarity-based spatial–spectral fusion hyperspectral image classification with deep CNN and Gabor filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514215.
[50] Q. Guo, S. Li, and A. Li, "An efficient dual spatial–spectral fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412913.
[51] Z. Qiu, J. Xu, J. Peng, and W. Sun, "Cross-channel dynamic spatial–spectral fusion transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528112.
[52] H. Li and X.-J. Wu, "CrossFuse: A novel cross attention mechanism based infrared and visible image fusion approach," *Inf. Fusion*, vol. 103, Mar. 2024, Art. no. 102147.
[53] J. Shen, Y. Chen, Y. Liu, X. Zuo, H. Fan, and W. Yang, "ICAFusion: Iterative cross-attention guided feature fusion for multispectral object detection," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109913.
[54] J. Song, C. Mou, S. Wang, S. Ma, and J. Zhang, "Optimization-inspired cross-attention transformer for compressive sensing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6174–6184.
[55] W. Ma, H. Zhang, M. Ma, C. Chen, and B. Hou, "ISSP-Net: An interactive spatial–spectral perception network for multimodal classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4412014.



Mengru Ma received the B.S. degree in communication engineering from Hebei University of Engineering, Handan, China, in 2019, and the Ph.D. degree from Xidian University, Xi'an, China, in 2024.

She is currently a Post-Doctoral Researcher with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University. Her research interests include deep learning, image interpretation, and multi-resolution remote sensing (RS) image fusion classification.



Jiaxuan Zhao (Graduate Student Member, IEEE) received the B.S. degree in materials science and engineering from Xidian University, Xi'an, China, in 2019, where she is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence.

Her research interests include remote sensing image understanding, multimodal fusion, and evolutionary computing.



Wenping Ma (Senior Member, IEEE) received the B.S. degree in computer science and technology and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

She is currently a Professor with the School of Artificial Intelligence, Xidian University. Her research interests include natural computing and intelligent image processing.

Dr. Ma is a member of the Chinese Institute of Electronics (CIE).



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiaotong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with Xidian University, Xi'an, where he is currently a Distinguished Professor and serves with the School of Artificial Intelligence. He currently serves as the Director of the Key Laboratory of Intelligent Perception and Image Understanding, which is affiliated with the Ministry of Education of China. His research interests include machine learning, deep learning, natural computation, remote sensing, image processing, and intelligent information processing.

Prof. Jiao has been a member of the Academia Europaea and a fellow of IEEE/IET/CAAI/CIE/CCF/CAA/CSIG/AAIA/AIIA/ACIS. He is the Chairperson of the Awards and Recognition Committee in IEEE Xi'an Branch, the sixth and seventh Vice Chairperson of the Chinese Association of Artificial Intelligence, the Chairperson of the Asian Society for Computational Intelligence, a councilor of the Chinese Institute of Electronics, a committee member of the Chinese Committee of Neural Networks, an expert of the Academic Degrees Committee of the State Council, and an ESI highly cited scientist.



Lingling Li (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2011 and 2017, respectively.

From 2013 to 2014, she was an Exchange Ph.D. Student with the Intelligent Systems Group, Department of Computer Science and Artificial Intelligence, University of the Basque Country UPV/EHU, Leioa, Spain. She is an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence, Xidian University. Her research interests include image processing, deep learning, and pattern recognition.

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Xu Liu (Senior Member, IEEE) received the B.S. degree in mathematics and applied mathematics from North University of China, Taiyuan, China, in 2013, and the Ph.D. degree from Xidian University, Xi'an, China, in 2019.

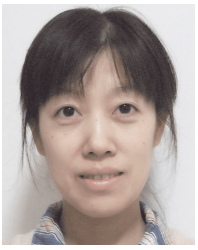
He is currently an Associate Professor of Huashan Elite and a Post-Doctoral Researcher with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University. His research interests include machine learning and image processing.



Fang Liu (Senior Member, IEEE) received the B.S. degree in computer science and technology from Xi'an Jiaotong University, Xi'an, China, in 1984, and the M.S. degree in computer science and technology from Xidian University, Xi'an, in 1995.

She is currently a Professor with Xidian University. She has authored or co-authored five books and more than 80 articles in journals and conferences. Her research interests include image perception and pattern recognition, machine learning, evolutionary computation, and data mining.

Prof. Liu won the second prize of the National Natural Science Award in 2013.



Shuyuan Yang (Senior Member, IEEE) received the B.A. degree in electrical engineering, and the M.S. and Ph.D. degrees in circuit and system from Xidian University, Xi'an, China, in 2000, 2003, and 2005, respectively.

She has been a Professor with the School of Artificial Intelligence, Xidian University. Her research interests include machine learning and multiscale geometric analysis.

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